

MURUGAN M B

Java Developer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Professional Summary

B.Sc. Computer Science graduate with hands-on experience building production-grade Java/Spring Boot applications and Python-based ML systems. Developed Scheme Connect, a full-stack REST API web application using Spring Boot, Hibernate/JPA, and MySQL with 5+ endpoints and real-time eligibility-based recommendations. Built Garden Guardian, an AI plant disease detection system achieving 98.6% accuracy across 38+ categories on 10,000+ images, deployed live on Vercel. Strong fundamentals in OOP, Collections, Multithreading, JDBC, and Exception Handling. Certified in Java (NPTEL – IIT Kanpur, in progress) and SQL (HackerRank). Seeking a Java Developer role to build scalable, production-grade backend systems.

Technical Skills

Languages	:	Java (primary), Python
Core Java	:	OOP, Collections Framework, Multithreading, Exception Handling, JDBC, Java 8+ (Streams, Lambdas)
Backend	:	Spring Boot, Spring MVC, Spring REST, Hibernate / JPA, Maven
Databases	:	MySQL, SQL — query optimisation, schema design
Testing & Tools	:	JUnit (unit testing), Postman (API testing), Git, GitHub, VS Code
Web	:	HTML5, CSS3 (front-end integration)
AI / ML	:	Data Preprocessing, Feature Engineering, Model Training & Evaluation (Python)

Internship Experience

Python with AI Intern | Infosmite

2025

- Built a wine quality prediction ML model on 5,000+ records, achieving 85–92% accuracy through iterative feature engineering and hyperparameter tuning.
- Executed the complete ML pipeline independently: data collection, preprocessing, model training, cross-validation, evaluation, and reporting.
- Applied Python best practices throughout the project lifecycle; delivered all milestones on schedule.

Projects

Scheme Connect · Java · Spring Boot · REST API · MySQL · HTML · CSS

Government scheme recommendation system · [GitHub Repository](#)

- Designed and developed a full-stack Spring Boot + MVC web application recommending government schemes based on user eligibility, served via 5+ RESTful API endpoints.

- Implemented global exception handling and proper request/response lifecycle management, ensuring consistent error responses across all endpoints.
- Optimised MySQL queries using Hibernate ORM and JDBC, reducing query complexity for real-time eligibility-based data retrieval.
- Followed end-to-end SDLC: requirement analysis → modular design → development → API testing with Postman.

Garden Guardian · Python · Machine Learning · HTML · CSS

AI plant disease detection system · [GitHub Repository](#) · [Live Demo](#)

- Trained a CNN-based image classification model on 10,000+ plant images, achieving 98.6% accuracy across 38+ disease categories.
- Delivered real-time disease diagnosis in under 2 seconds with actionable treatment and prevention recommendations per detected disease.
- Built end-to-end application pipeline: data preprocessing → model training → web interface integration → live deployment on Vercel.

Wine Quality Prediction · Python · Scikit-learn · Pandas · Machine Learning

- Developed a predictive model on 5,000+ wine quality records achieving 85–92% accuracy through feature engineering, cross-validation, and hyperparameter tuning.
- Applied full ML workflow: data preprocessing → feature selection → model training → performance evaluation and reporting.

Certifications

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- Java Programming — NPTEL (IIT Kanpur) · Certification in progress
 - SQL (Advanced) — HackerRank · Verified badge

Education

B.Sc. Computer Science Sadakathullah Appa College	2023 – 2026
CGPA: 6.94 / 10 (from 1 st semester - 5 th semester) · Tirunelveli, Tamil Nadu	
Higher Secondary (HSC) Schaffter Higher Secondary School	2023
SSLC Schaffter Higher Secondary School	2021

Achievements

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- Tamil Nadu Government Startup Idea Competition — Selected participant among state-level applicants.
 - State-Level Symposium (Web Design) — Presented project to a panel of industry judges.
 - State-Level Symposium (Logo Design) — Showcased design work and received formal recognition.